Main Line Belt Conveyor (CONV-BELT) Equipment Manual and SOP

VINDHYA PRECISION WORKS ## EQUIPMENT MANUAL AND MAINTENANCE STANDARD OPERATING PROCEDURE ### ASSET CLASS: CONV-BELT — MAIN LINE BELT CONVEYOR
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Effective Date: 14 October 2024 Applies To: Plant Maintenance Department, Building 4 Heavy Assembly Line

1. PURPOSE AND SCOPE

This document defines the technical specifications, operating ranges, preventive maintenance schedules, fault-clearing procedures, and spare parts catalog for the Main Line Belt Conveyor, asset class CONV-BELT.

Adherence to this Standard Operating Procedure (SOP) is mandatory for all maintenance technicians, reliability engineers, and operations personnel at Vindhya Precision Works.

2. RESPONSIBLE TRADES AND HAZARDOUS ENERGY ISOLATION

2.1 Primary Trade * Mechanical Technician: Responsible for all mechanical inspections, alignments, roller replacements, belt tensioning, and gearbox servicing.

2.2 Supporting Trade * Electrical Technician: Required to assist during lockout/tagout, drive motor checks, sensor recalibrations, and electrical control troubleshooting.

2.3 Hazardous Energy Sources Prior to commencing any maintenance activity outside of standard visual inspection, technicians must isolate the following energy sources in accordance with Vindhya Precision Works Lockout/Tagout (LOTO) Procedure VPW-LOTO-01: 1. Electrical: 415V AC 3-Phase supply via Motor Control Center (MCC) Panel B4-MCC-07, Feeder Breaker 12. 2. Mechanical: Stored rotational energy of the head pulley and gravity take-up weight assembly; mechanical brake engagement required.

3. NORMAL OPERATING RANGES AND CONDITION-MONITORING THRESHOLDS

The condition-monitoring system (CMS) monitors asset class CONV-BELT continuously. Healthy equipment operating within baseline parameters exhibits a vibration velocity of 1.2 to 4.0 mm/s RMS and a surface temperature of 38 to 66 °C.

The specific numeric operating limits, alarm setpoints, and trip thresholds for asset class CONV-BELT are defined in Table 1.

Table 1: Operating Limits and Alarm Setpoints (CONV-BELT)

Parameter	Normal Healthy Range	Warning Setpoint	Trip Setpoint	Action on Trip
Vibration (Velocity RMS)	1.2 – 4.0 mm/s	7.1 mm/s RMS	11.0 mm/s RMS	Immediate drive interlock shutdown
Surface Temperature (Bearings/Drive)	38 – 66 °C	70 °C	84 °C	Immediate drive interlock shutdown
Motor Current (FLA 45A nominal)	32.0 – 42.0 A	46.5 A (105% FLA)	49.5 A (110% FLA for >5s)	Overload relay trip and lockout
Process Pressure	N/A	N/A	N/A	*No monitored process pressure on this asset class.*

4. PREVENTIVE MAINTENANCE SCHEDULE

The base preventive maintenance (PM) interval for asset class CONV-BELT is 1000 running hours (± 48 operational hours). Maintenance tasks are categorized by cumulative running hours in Table 2.

Table 2: Preventive Maintenance Running-Hour Schedule

Interval	Task ID	Maintenance Task Description	Trade
1,000 Hours	PM-1K-01	Inspect belt tracking, alignment, and edge wear. Check skirt rubber clearance.	Mechanical
1,000 Hours	PM-1K-02	Grease head and tail pulley plummer block bearings (Lithium Complex NLGI 2, 40g per point).	Mechanical
1,000 Hours	PM-1K-03	Inspect drive coupling alignment and fastener torque.	Mechanical
1,000 Hours	PM-1K-04	Clean optical and inductive proximity sensors. Verify signal response.	Electrical
3,000 Hours	PM-3K-01	Check gearbox oil level, condition, and breathers. Take oil sample for laboratory analysis.	Mechanical
3,000 Hours	PM-3K-02	Perform thermal imaging scan of motor windings, contactors, and bearing housings.	Electrical
6,000 Hours	PM-6K-01	Complete overhaul and wear check of impact idlers and return rollers. Replace failed units.	Mechanical
6,000 Hours	PM-6K-02	Megger test drive motor windings (minimum acceptable insulation resistance: 100 MΩ at 500V DC).	Electrical
12,000 Hours	PM-12K-01	Flush and replace gearbox synthetic lubricant (ISO VG 220 gear oil).	Mechanical
12,000 Hours	PM-12K-02	Inspect main drive contactor contacts and replace if pitting exceeds 1.5 mm depth.	Electrical

5. FAULT-CODE AND TROUBLESHOOTING TABLE

When the control system flags an operational fault, technicians shall reference Table 3 for root-cause diagnosis and corrective action.

Table 3: Fault-Code Descriptions, Causes, and Corrective Actions

Fault ID	Description (Prose)	Probable Cause	Corrective Action
F-CB-01	Drive Contactor Failure to Engage	Internal coil burnout or mechanical binding in starter assembly.	Isolate power; inspect and replace the Attack contactor (Part VPW-P-00016).
F-CB-02	Bearing Lubrication / Seal Integrity Failure	Grease contamination ingress past worn shaft barriers.	Isolate power; replace degraded barrier component using Avoid seal (Part VPW-P-00025).
F-CB-03	Speed/Motion Monitoring Loss	Optical/magnetic pulse train interruption from tracking unit.	Clean lens; if unit is non-responsive, replace with Artist sensor (Part VPW-P-00028).
F-CB-04	Pulley Shaft End-Cap Leakage	Lubricant seepage past secondary housing exclusion barriers.	Depressurize and overhaul housing; install new Drop seal (Part VPW-P-00029).
F-CB-05	Longitudinal Carcass Delamination or Tear	Mechanical snagging of conveyed product against structural stringers.	Isolate asset; cut out damaged section and splice in a replacement Avoid belt (Part VPW-P-00035).
F-CB-06	Auxiliary Control Circuit Dropout	Chatter and failure of secondary auxiliary switching interface.	Verify control voltage; replace faulty auxiliary relay using Mention contactor (Part VPW-P-00040).
F-CB-07	Misalignment Edge-Switch Trip	Lateral belt drift exceeding allowable track envelope limits.	Adjust tracking idlers; check sensor alignment and replace faulty unit with Several sensor (Part VPW-P-00044).
F-CB-08	Return Roller Bearing Seizure	Internal ball-race corrosion or particulate blinding.	Lock out conveyor; extract seized assembly and fit a replacement Own seal (Part VPW-P-00047) inclusive of roller cartridge.

6. APPROVED SPARE PARTS LIST

Technicians must strictly use plant-inventory-controlled part numbers listed in Table 4 for all replacement work on asset class CONV-BELT. No substitutions are permitted without written sign-off from the Plant Reliability Manager.

Table 4: Approved Spare Parts Inventory

Part Number	Plant Inventory Description	Asset Application
VPW-P-00016	Attack contactor	Main motor starter primary power switching
VPW-P-00025	Avoid seal	Primary shaft bearing contamination barrier
VPW-P-00028	Artist sensor	Shaft rotational speed and zero-speed monitoring
VPW-P-00029	Drop seal	Secondary housing lubrication retention barrier
VPW-P-00035	Avoid belt	Main conveyor carcass replacement section
VPW-P-00040	Mention contactor	Auxiliary control circuit switching relay
VPW-P-00044	Several sensor	Edge-tracking and belt-sway proximity detector
VPW-P-00047	Own seal	Return roller and idler bearing protection assembly

7. TORQUE SPECIFICATIONS

To prevent structural failure or fastener fatigue during maintenance assembly, all technicians must apply calibrated torque wrenches to the fastener classes detailed in Table 5.

Table 5: Fastener Torque Guidelines

Fastener Location / Size	Thread Class	Target Torque (Nm)	Tolerance
Drive Motor Mounting Bolts (M16)	Grade 8.8	185 Nm	± 5 Nm
Pillow Block Bearing Holding Down Bolts (M20)	Grade 8.8	360 Nm	± 10 Nm
Head Pulley Locking Assembly Screws (M12)	Grade 10.9	115 Nm	± 3 Nm
Conveyor Stringer Structural Splice Bolts (M12)	Grade 8.8	85 Nm	± 2 Nm
Gearbox Torque Arm Anchor Bolts (M24)	Grade 10.9	650 Nm	± 15 Nm

8. SAFETY-CRITICAL CORRECTIVE ACTIONS AND PERMIT REQUIREMENTS

The following corrective actions involve high-risk energy states or structural load manipulations. They are explicitly designated as Safety-Critical.

Before initiating any of the tasks listed below, technicians must execute a formal Lockout/Tagout (LOTO) per Procedure VPW-LOTO-01 AND obtain an active Permit to Work (PTW) issued by the Shift Safety Officer:

1. Replacement of Main Drive Contactor (Fault F-CB-01 / Part VPW-P-00016): Involves live-side 415V exposure risk within the MCC enclosure. LOTO of MCC Feeder Breaker 12 and High-Voltage PTW mandatory.
2. Conveyor Belt Splice and Replacement Operations (Fault F-CB-05 / Part VPW-P-00035): Involves release and re-tensioning of the gravity take-up weight stack (stored potential energy) and heavy manual handling. Mechanical LOTO, gravity drop-stop pin installation, and Hot Work / Lifting PTW mandatory.
3. Head Pulley and Bearing Assembly Overhaul (Fault F-CB-02 / Part VPW-P-00025): Involves overhead rigging, lifting of heavy rotating assemblies (>250 kg), and decoupling of heavy-duty gearboxes. Rigging Plan, Craning Permit, and Mechanical LOTO mandatory.

End of Document — Vindhya Precision Works Internal Policy